1.8.4 Metal Oxide Resistors

It is manufactured by coating tin oxide and antimony on a glass strip. It withstands high temperature and moisture. It is rolled and cut to get desired value. Figure 1.33 shows a metal oxide resistor

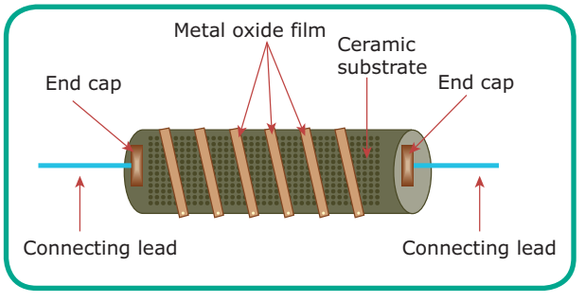


Figure 1.33 Metal oxide resistor

Special Features

1. It has long life
2. It operates at high frequency
3. It is not affected by moisture
4. It can withstand temperature up to 450°C

1.8.5 Metal Film Resistors

There are two types of metal film resistors, viz. Thick film resistor and thin film resistor.

1.8.5.1 Thick Film Resistor

This type of resistor is in the form of cylinder (or) plate shape. Inside the cylinder, there is nichrome metal film as shown in Figure 1.34. It has more physical strength.

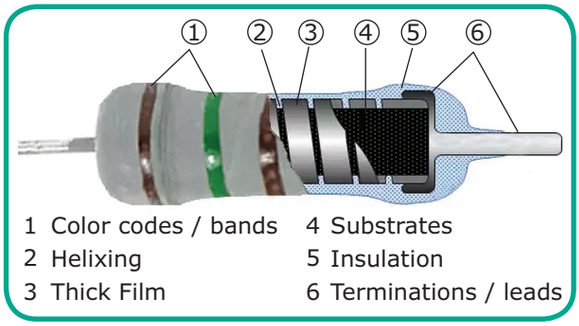


Figure 1.34 Cross-sectional view of thick film resistor

Uses

1. Used in high frequency circuits such as mobile phone transmitters.
2. Used in computers.
3. Used in motor controlled circuits.
4. Used in radar systems.

1.8.5.2 Thin Film Resistor

This type of resistor is manufactured on high quality ceramic (or) glass tube in which nichrome metallic vapour is coated. It was heated to high temperature and slowly cooled down (called annealing). Depending on the thickness and materials of the thin film, desired values of resistors can be produced. Figure 1.35 shows a thin film resistor.

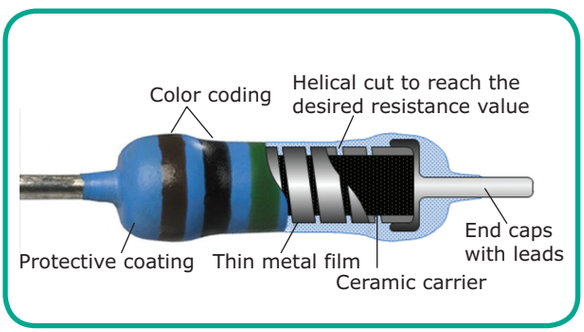


Figure 1.35 Cross sectional view of the thin film resistor

1.8.6 Printed Resistors

A carbon resistor which is printed directly on a Printed Circuit Board (PCB) is called as printed resistor. It is shown in Figure 1.36.

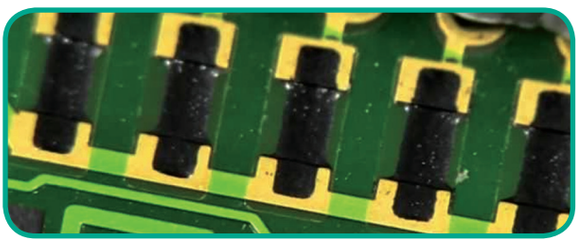


Figure 1.36 Printed resistor

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